

## **ITA1407 – Ethical Hacking**

### **CO3 - Assessment Tool 3 – BTL (Analyze)**

## **Penetration Testing – Analyze**

### **Experiment: Analyze Nmap Port Scan Results**

#### **Aim:**

To analyze the ports of an authorized target system using Nmap and identify the meaning of Open, Closed, and Filtered port states.

#### **Objective:**

- To identify open ports on the target.
- To identify closed ports.
- To determine whether ports are filtered by a firewall.
- To understand the security implications of exposed services.

#### **Tool Used:**

Nmap – Network Mapper

#### **Target:**

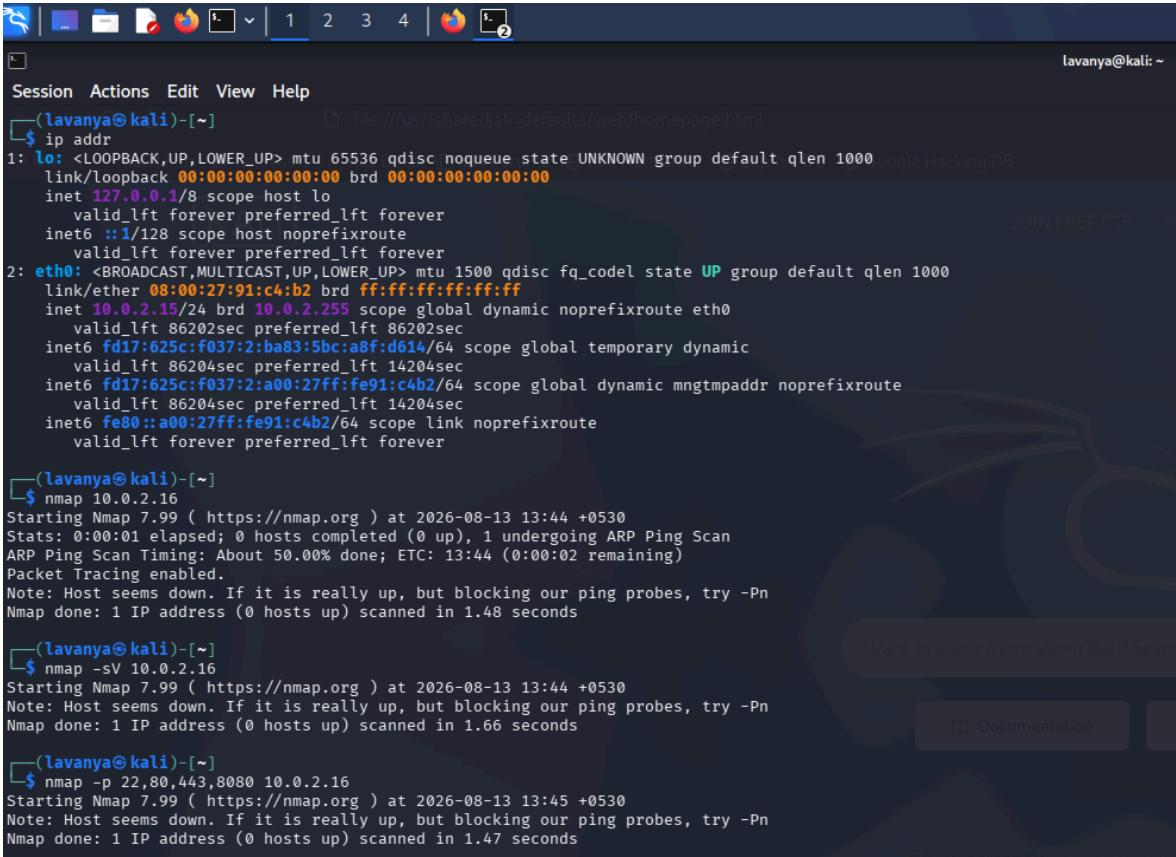
Authorized lab system / Virtual Machine

#### **Procedure**

1. Identify the IP address of the authorized target machine.
2. Check whether the target is reachable.
3. Perform a basic Nmap port scan.
4. Perform service/version detection.
5. Analyze the reported port states.
6. Record the security implications of the findings.

#### **Commands**

- Ip addr
- nmap 10.0.2.16
- nmap -sV 10.0.2.16
- nmap -p 22,80,443,8080 10.0.2.16



```
(lavanya@kali)-[~]
$ ip addr
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
        valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host noprefixroute
        valid_lft forever preferred_lft forever
2: eth0: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
    link/ether 08:00:27:91:c4:b2 brd ff:ff:ff:ff:ff:ff
    inet 10.0.2.15/24 brd 10.0.2.255 scope global dynamic noprefixroute eth0
        valid_lft 86202sec preferred_lft 86202sec
    inet6 fd17:625c:f037:2:ba83:5bc:a8f:d614/64 scope global temporary dynamic
        valid_lft 86204sec preferred_lft 14204sec
    inet6 fd17:625c:f037:2:a00:27ff:fe91:c4b2/64 scope global dynamic mngtmpaddr noprefixroute
        valid_lft 86204sec preferred_lft 14204sec
    inet6 fe80::a00:27ff:fe91:c4b2/64 scope link noprefixroute
        valid_lft forever preferred_lft forever

(lavanya@kali)-[~]
$ nmap 10.0.2.16
Starting Nmap 7.99 ( https://nmap.org ) at 2026-08-13 13:44 +0530
Stats: 0:00:01 elapsed; 0 hosts completed (0 up), 1 undergoing ARP Ping Scan
ARP Ping Scan Timing: About 50.00% done; ETC: 13:44 (0:00:02 remaining)
Packet Tracing enabled.
Note: Host seems down. If it is really up, but blocking our ping probes, try -Pn
Nmap done: 1 IP address (0 hosts up) scanned in 1.48 seconds

(lavanya@kali)-[~]
$ nmap -sV 10.0.2.16
Starting Nmap 7.99 ( https://nmap.org ) at 2026-08-13 13:44 +0530
Note: Host seems down. If it is really up, but blocking our ping probes, try -Pn
Nmap done: 1 IP address (0 hosts up) scanned in 1.66 seconds

(lavanya@kali)-[~]
$ nmap -p 22,80,443,8080 10.0.2.16
Starting Nmap 7.99 ( https://nmap.org ) at 2026-08-13 13:45 +0530
Note: Host seems down. If it is really up, but blocking our ping probes, try -Pn
Nmap done: 1 IP address (0 hosts up) scanned in 1.47 seconds
```

Observation

Record the output in the following format:

| Port | State  | Service | Observation                                   |
|------|--------|---------|-----------------------------------------------|
| XX   | Open   | XXXX    | Service is running                            |
| XX   | Closed | XXXX    | Host is reachable but no service is listening |

|    |         |      |                                      |
|----|---------|------|--------------------------------------|
| XX | Filtere | XXXX | Firewall/filter prevents determining |
|    | d       | X    | the state                            |

## **Analysis**

### **Open:**

An open port indicates that an application or service is actively listening on that port. Such services should be checked to determine whether they are necessary and securely configured.

### **Closed:**

A closed port indicates that the target host is reachable, but no application is currently listening on that port.

### **Filtered:**

A filtered port indicates that a firewall, ACL, or other network filtering mechanism is preventing Nmap from determining whether the port is open.

## **Security Assessment**

The open ports identified during the scan represent potential attack surfaces. Unnecessary services should be disabled, while required services should be properly configured and updated. Filtered ports indicate that network filtering controls may be in place.

## **Result**

The Nmap scan successfully identified the port states of the authorized target. The results were analyzed to distinguish between Open, Closed, and Filtered ports and to identify potential network attack surfaces.